

Rederived Closed-Form Continuous-Time Neural Networks

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Abstract—In the field of machine learning, continuous-time neural networks (CTNNs) have garnered significant attention due to their powerful modeling capabilities for time series data. However, traditional numerical ordinary differential equation (ODE) solving methods suffer from computational inefficiency. Although closed-form CTNNs (CfCs) alleviate this issue through closed-form approximate solutions, their zero-order approximation method based on segmented constant inputs still exhibits errors when processing high-resolution or irregularly sampled data. This study proposes a numerical integration approximation theory based on the Lagrange interpolation method (NALIM), which introduces temporal interval factors to linearly approximate the nonlinear integral function in liquid time-constant networks (LTCs) network state equations, deriving closed-form solutions dependent on external inputs and sampling intervals, while rigorously proving the error upper bound between the approximate and original solutions. Furthermore, we construct two novel network architectures: the dynamic feature accumulation closed-form network (DFA-CfN) for general sequence tasks, and the prereconstructed DFA-CfN (PRDFA-CfN) optimized for decomposable sequence tasks, along with six variant networks from these two families. Extensive experiments across six distinct tasks—physical dynamics modeling, human activity recognition, PhysioNet challenge, internet movie database (IMDb) sentiment analysis, XCO₂ remote sensing sequence prediction, and multienergy load forecasting in integrated energy systems (IESs)—demonstrate that our proposed models achieve state-of-the-art performance across all benchmarks, with the highest metrics improvement reaching 69.2%, providing a novel high-performance tool for continuous-time modeling tasks. The code used to support the findings of this study is available from <https://github.com/nuist-xf/DFA-CfNs>

Index Terms—Continuous-time neural networks (CTNNs), dynamic feature accumulation, liquid networks, numerical integration approximation theory, time series reconstruction.

I. INTRODUCTION

IN THE field of machine learning, continuous innovation in neural network architectures aims to address the

challenges of complex data processing and enhance efficiency. Continuous-time neural networks (CTNNs) defined by ordinary differential equations (ODEs), which possess continuous-time hidden layer states, represent an effective method for modeling time series data widely present in healthcare, industrial, commercial, and other domains.

Since the advent of liquid time-constant networks (LTCs) [1], significant improvements have been demonstrated in subsequent works [2], [3], [4]. However, similar to traditional CTNNs, their reliance on numerical ODE solvers introduces inherent computational inefficiencies. The research community has proposed numerous enhancements to address this limitation [5], [6], [7], [8], [9], [10], [11], [12]. Notably, Closed-form CTNNs (CfCs) [12] pioneered an approximate closed-form solution for the state differential equations of LTC. By constructing networks based on this approximation, they achieved substantial reductions in inference time.

In real-world scenarios, many datasets cannot be consistently collected in continuous form. When temporal sampling resolution increases or becomes irregular, the iterative relationships between states may exhibit anomalous variations. In this case, the network that relies on zero-order approximations obtained from segmented constant inputs is unable to detect these abnormal patterns. Inspired by CfCs, this study proposes a numerical integration approximation theory based on the Lagrange interpolation method. This method introduces a time interval factor to perform a first-order approximation of the nonlinear integral function in the LTC state equation, thereby obtaining an approximate solution form dependent on exogenous inputs and sampling intervals. In addition, we rigorously prove the upper bound error between the approximate solution and the original solution. Notably, as the sampling interval approaches infinitesimal, our solution converges to the same form as the foundational CfCs. Building on this rederived approximation, we incorporated the mechanism of dynamic correction at each time step into the neural network and implemented forward feature accumulation to enhance feature modeling based on the RNN structure. We proposed two new neural network series for explicit optimization of real-world discrete sequence inputs. The constructed model series were comprehensively tested on diverse tasks, including four fundamental continuous time series modeling tasks [physical dynamics modeling, human activity recognition, PhysioNet challenge, sentiment analysis using internet movie database (IMDb)] and two newly constructed engi-

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neering time series tasks [*XCO₂ remote sensing sequence prediction, multienergy loads forecasting for integrated energy systems (IESs)*], consistently achieving state-of-the-art performance across all benchmarks. The principal contributions of this work are as follows.

- 1) A rederived closed-form approximate solution for liquid time-constant state equations, with a rigorous error upper bound proof, establishes novel analytical methods.
- 2) The constructed numerical integration approximation theory based on the Lagrangian interpolation method for state function approximation ensures that the approximate solutions comprehensively incorporate the influence of sampling intervals on real-world sequential characteristics.
- 3) Two novel neural network architectures constructed from the proposed solution as follows.
 - a) *Dynamic Feature Accumulation Closed-form Network*: This is optimized for general sequential tasks.
 - b) *Prereconstruct DFA-CfN*: This is specialized for decomposable sequential input tasks.
- 4) Comprehensive benchmarking across six domains demonstrates that the proposed models achieve state-of-the-art performance on all evaluated datasets.

The dynamic feature accumulation closed-form network (DFA-CfN) is based on the numerical integration approximation theory based on Lagrange interpolation method (NALIM) theory, which dynamically corrects the features at each time step, while the RNN structure makes it possible to accumulate historical state features for more accurate feature modeling. Prereconstruct DFA-CfN (PRDFA-CfN) adds a pre-reconstruction (PR) mechanism for decomposable sequences or periodic sequences on the basis of DFA-CfN, which is able to synchronously enhance the periodicity of the sequences and de-realization, and the mechanism can be used alone as a forward feature enhancer before any kind of temporal modeling network.

II. RELATED WORKS

Traditional neural networks rely on discrete hidden layers to transform information, whereas the state of a neural ODE [13] is defined by the solution to the following equation:

$$dx(t)/dt = f(x(t), I(t), t, \theta) \quad (1)$$

where $x(t)$ represents the hidden state, $I(t)$ is an exogenous input to the system, and the neural network function f is parameterized by θ . Starting from the initial state $x(0)$, the output $x(T)$ is obtained via a numerical ODE solver, while the adjoint sensitivity method [14] is used to address the challenge of backpropagation through the ODE solver. This architecture has shown potential in tasks such as trajectory modeling [15], medical image analysis [16], and modeling sequential and irregularly sampled data [17]. Researchers have proposed various strategies to reduce the computational cost of ODE solvers, such as state augmentation techniques [5], [6], reformulating forward propagation as a root-finding problem [7], regularization schemes [8], [9], [10], and hyper-solver designs [11]. However, the training and inference processes still face speed bottlenecks due to the reliance on numerical

solvers, with significant computational overhead in complex data, tasks, and state space scenarios.

Inspired by the dynamics of biological neurons, LTCs [1] emerged as a new instance of RNNs, drawing on the modeling ideas of the dynamic changes in neuron potentials [18] and the concepts of dynamic causal models [19], forming their unique network structure. The hidden state of LTCs is modulated by a linear first-order dynamical system and nonlinear gating, introducing time constants τ that can dynamically change with the input in the differential equation. This method achieves more accurate modeling by introducing constants (liquid time constants) that adaptively change over time. The dynamical equation is

$$dx(t)/dt = -[\omega_\tau + f(x, I, \theta)] \odot x(t) + A \odot f(x, I, \theta). \quad (2)$$

Theoretically, LTCs possess stable dynamics and strong expressive power, outperforming traditional and modern recurrent neural networks in several time series prediction tasks [1]. However, LTCs are still limited by numerical solvers, and their training and inference speeds need improvement.

Subsequently, CfCs [12] made significant advancements. CfCs achieve an approximate closed-form solution to the ODE by tightly approximating the integral of neuron and synapse interactions in the LTC network. Specifically, CfC applies the theory of linear ODEs to analytically solve for the dynamics of the LTC DE, and then, CfC reduces the solution to just one integral and discretizes $I(s)$ into discrete constant segments and obtains a discrete approximation of the integral by summing over the discrete constant intervals. For the initial value problem (IVP) of a single neuron LTC system in (2), through linear ODE theory and simplification, and by approximating the exogenous input in the integral term $\int_0^t f(I(s))ds$ (f satisfies certain conditions) as piecewise constants, an approximate closed-form solution is obtained

$$x(t) \approx (x_0 - A) e^{-[\omega_\tau + f(I(t), x(t), \theta)]t} f(-I(t), x(t), \theta) + A. \quad (3)$$

Thus, for any point in continuous time, there is a corresponding approximate solution, eliminating the dependence on complex numerical solvers. This significantly improves training and inference speeds compared to ODE-based models, providing a more advantageous tool for solving complex problems in practical applications, and demonstrating superior performance in multiple tasks [20], [21], [22], [23].

But for the approximation of piecewise constant input $I(t)$, and f is monotonically increasing, bounded, nonnegative, and continuous, then in $f(I(t))$, the trend of f with respect to t must be the same as that of I with respect to t , the trend before and after the piecewise constant approximation can be regarded as Fig. 1. Obviously, this approximation has some errors, and because the exogenous input I tends to be variable, the approximation error will increase with the increase of the sampling resolution. A potential reason is that this approximation ignores the long-term trend of the function f relative to t . When the resolution changes, the function value on the interval can only be replaced by a constant, and the influence of the interval length cannot be considered. Therefore, it is necessary to establish a new approximation method that can comprehensively consider the function and

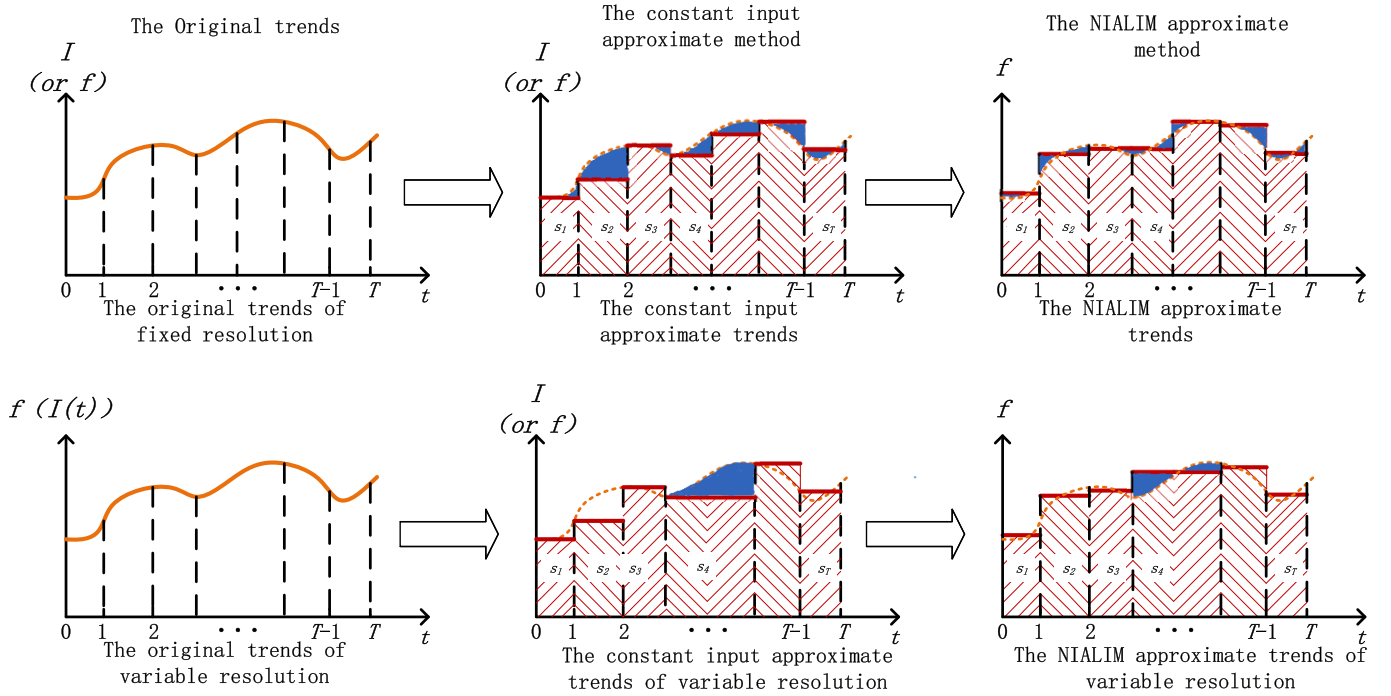


Fig. 1. Piecewise constant inputs (approximation methods used in CfC) and NIALIM approximate I , f , and $\int_0^t f(s)ds$, which is equal to the sum of the areas of the rectangles. The blue area is the approximation error, which can be effectively reduced by the NIALIM method compared to the segmented constant input because it will adjust the function value of the starting point by linear weights, so that the approximation error is almost minimized.

interval characteristics to reduce the approximation error in this process.

III. THEORIES

The hidden state of an LTC network is determined by the solution of the following IVP:

$$\frac{d\mathbf{x}}{dt} = -[\omega_\tau + f(\mathbf{x}, \mathbf{I}, \theta)] \odot \mathbf{x}(t) + A \odot f(\mathbf{x}, I, \theta) \quad (4)$$

where at a time step t , $\mathbf{x}^{(D \times 1)}(t)$ defines the hidden state of a LTC layer with D cells, and $\mathbf{I}^{(m \times 1)}(t)$ is an exogenous input to the system with m features. Here, $\omega_\tau^{(D \times 1)}$ is a time-constant parameter vector, $A^{(D \times 1)}$ is a bias vector, f is a neural network parametrized by θ and $\odot$ is the Hadamard product. The dependence of $f(\cdot)$ on $\mathbf{x}(t)$ denotes the possibility of having recurrent connections.

A. Lagrange Interpolation Method

According to [24], given $n + 1$ distinct nodes

$$\{(x_i, y_i)\}_{i=0}^n \subset [a, b] \quad (5)$$

and their function values

$$\{y_i = f(x_i)\}_{i=0}^n \quad (6)$$

we seek a unique polynomial $L_n(x) \in \mathbf{P}_n$ that satisfies the interpolation conditions

$$L_n(x_i) = y_i \quad \forall i = 0, 1, \dots, n. \quad (7)$$

First, we define the Lagrange basis functions $l_i(x_j)$ which satisfy the Kronecker delta function

$$l_i(x_j) = \delta_{ij} = \begin{cases} 1, & j = i \\ 0, & j \neq i \end{cases} \quad (8)$$

where the explicit expression is

$$l_i(x) = \prod_{\substack{m=0 \\ m \neq i}}^n \frac{x - x_m}{x_i - x_m}, \quad i = 0, 1, \dots, n \quad (9)$$

where x_m are the interpolation nodes and m is the node index. The numerator $x - x_m$ ensures that $l_i(x)$ is zero when $x - x_m (m \neq i)$. The denominator $x_i - x_m$ is a normalization factor that ensures $l_i(x_i) = 1$. Each $l_i(x)$ is an n -degree polynomial and satisfies orthogonality at the nodes. The interpolation polynomial is constructed through a linear combination of the basis functions

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (10)$$

where the linear combination coefficients y_i directly ensure $L_n(x_i) = y_i$. If the interpolated function is $f(x) \in C^{n+1}[a, b]$, then for any $x \in [a, b]$, there exists a $\xi \in (a, b)$ in the interval such that

$$f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i) \quad (11)$$

where $f^{(n+1)}(\xi)$ is the $(n + 1)$ th derivative of f at some point ξ within the interval. The term $\prod_{i=0}^n (x - x_i)$ is the node

polynomial, whose roots are located at all the interpolation nodes.

B. Numerical Integration Approximation Theory Based on Lagrange Interpolation Method (NALIM)

This section will use the Lagrange method obtained in Section III-A to approximate the integrand.

First, consider the case of a general integral discretization representation. Let the function $f(x)$ be continuous (or integrable) on the interval $[a, b]$, to compute its definite integral

$$I = \int_a^b f(x) dx \quad (12)$$

the core idea of numerical integration is to approximate the integral using function values at a finite number of discrete points. The general form is

$$I \approx \sum_{j=1}^n \omega_j f(x_j) \quad (13)$$

where $\{x_j\}_{j=1}^n$ are the integration nodes distributed within the interval $[a, b]$, and $\{\omega_j\}_{j=1}^n$ are the corresponding weights.

This weight is generally passively fit according to the data in neural networks. This study uses the Lagrange method to obtain the expression form of this weight. Construct the Lagrange interpolation polynomial $P_{n-1}(x)$ that approximates $f(x)$ at the nodes $\{x_j\}_{j=1}^n$

$$P_{n-1}(x) = \sum_{j=1}^n f(x_j) l_j(x) \quad (14)$$

substitute this polynomial into the integral expression

$$\int_a^b f(x) dx \approx \int_a^b P_{n-1}(x) dx = \sum_{j=1}^n f(x_j) \int_a^b l_j(x) dx \quad (15)$$

where $\omega_j = \int_a^b l_j(x) dx$ are the integrals of the basis functions over the interval $[a, b]$, which depend only on the positions of the nodes x_i and are independent of $f(x)$.

C. Theorem 1

Given an LTC system determined by the IVP in (4), constructed by one cell, receiving a single-dimensional time-series exogenous input $I(t)$ with no self-connections, the following expression is an approximation of its closed-form solution:

$$x(t) = (x_0 - A) e^{-\omega_\tau t - \omega_{t_s} \sum_{j=1}^n f(I(j), x(j), \theta)} + A \quad (16)$$

where $\omega_{t_s}^{(1 \times 1)}$ is an interval correction factor, which can be expressed as follows:

$$\omega_{t_s} = e^{t_s(\ln t_s - 1)}. \quad (17)$$

In particular, when the exogenous input $I(t_i)$ is observable at infinite time points, (16) is formulated as a closed-form continuous approximation of the solution consistent with CFC in the following form:

$$x(t) = (x_0 - A) e^{-[\omega_\tau + f(I(t), x(t), \theta)]t} + A \quad (18)$$

D. Proof of Theorem 1

1) *Integral Conversion Based on NIALIM Method:* For (4), we can obtain the following form after some simplifications (The detailed reasoning process is shown in (5) to (6) of the CfCs' article [12].)

$$x(t) = (x_0 - A) e^{-\omega_\tau t - \int_0^t f(I(s), x(s), \theta) ds} + A \quad (19)$$

where the integral term $\int_0^t f(I(s), x(s), \theta) ds$ cannot be solved directly due to the arbitrary form of the exogenous input $I(t)$ and the nonlinear nature of $f(\cdot)$. However, the nonlinear integral term can be approximated using the NIALIM method in the following form based on discrete function-value weighted summation:

$$\int_0^t f(I(s), x(s), \theta) ds \approx \sum_{i=1}^n \omega_i f(I(t_i), x(t_i), \theta) \quad (20)$$

where ω_i is the corresponding weight of the function value f at t_i . Thus, (4) morphs into

$$x(t) \approx (x_0 - A) e^{-\omega_\tau t - \sum_{i=1}^n \omega_i f(I(t_i), x(t_i), \theta)} + A \quad (21)$$

Now we need to derive the weight values. The integration interval $[0, t]$ is uniformly divided into n subintervals $[t_{i-1}, t_i]$, the length of each subinterval is t_s , then the original integral can be divided into the sum of several subinterval integrals

$$\int_0^t f(I(s), x(s), \theta) ds = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} f(I(s), x(s), \theta) ds \quad (22)$$

on each subinterval, the Lagrange interpolation polynomial method is applied to f (Due to the linewidth in the article, $f(I(s), x(s), \theta)$ is abstracted here as a self-contained function of $x(s)$, $I(s)$ and θ), i.e.,

$$f_i(t) \approx p_i(t) = \sum_{j=1}^n f(t_j) \frac{\prod_{m \neq j} (t - t_m)}{\prod_{m \neq j} (t_j - t_m)} \quad (23)$$

where t_j is one of the sampling points, t_m is the other sampling points on the interval $[t_{i-1}, t_i]$, and t is all points on the interval $[t_{i-1}, t_i]$ except t_m . In particular, we set only one pair of sampling points. When a fixed point is taken at t_i , then t_{i-1} is another sampling point t_m , so for integrals over each subinterval, the transformation can be

$$\int_{t_{i-1}}^{t_i} f(s) ds = \frac{f(t_i)}{t_s} \int_{t_{i-1}}^{t_i} \prod_{s \rightarrow t_{i-1}}^{s=t_i} (s - s_{i-1}) ds. \quad (24)$$

2) *Simplification of Cumulative Product Subdivisions:* For the cumulative term of (24), let $g(s)$ represent $s - s_{i-1}$, B represent the cumulative term, and take the natural logarithm of this term

$$\ln(B) = \sum_{s \rightarrow t_{i-1}}^{s=t_i} \ln(g(s)) \approx \int_{s \rightarrow t_{i-1}}^{t_i} \ln(g(s)) ds \quad (25)$$

for integrals of the form $\int_{s \rightarrow t_{i-1}}^{t_i} \ln(g(s))ds$, the final form can be easily obtained by using the method of integration by parts

$$\lim_{s \rightarrow t_{i-1}} \int_{s \rightarrow t_{i-1}}^{t_i} \ln(g(s))ds = t_s \ln t_s - t_s \quad (26)$$

thus B can be reduced to

$$B = \exp(t_s \ln t_s - t_s) \quad (27)$$

substituting into (24) yields

$$\int_{t_{i-1}}^{t_i} f(s)ds = f(t_i) \exp(t_s \ln t_s - t_s) \quad (28)$$

substituting into (22) yields

$$\int_0^t f(s)ds = \sum_{i=1}^n f(t_i) \exp(t_s \ln t_s - t_s) \quad (29)$$

thus $x(t)$ can be expressed as

$$\bar{x}_d(t) = (x_0 - A) e^{-\omega_r t - \omega_{t_s} \sum_{j=1}^n f(I(j), x(j), \theta)} f(-I(t), x(t), \theta) + A \quad (30)$$

when an infinite number of f points are known, n tends to infinity and t_s tends to 0. The original equation transforms to

$$x(t) = (x_0 - A) e^{-[\omega_r + f(I(t), x(t), \theta)]t} f(-I(t), x(t), \theta) + A. \quad (31)$$

3) *Error Bound Analysis:* The error bound

$$|x(t) - \bar{x}_d(t)| \leq |x_0 - A| e^{-\omega_r t} \quad (32)$$

is derived directly from the analytical forms of the exact solution $x(t)$ and approximate solution $\bar{x}_d(t)$

$$x(t) = (x_0 - A) e^{-\omega_r t - \int_0^t f(s)ds} + A \quad (33)$$

$$\bar{x}_d(t) = (x_0 - A) e^{-\omega_r t - \omega_{t_s} \sum_{j=1}^p f(t_j)} + A \quad (34)$$

where $\omega_{t_s} = \exp(t_s \ln t_s - 1)$ and t_s is the uniform sampling interval. The absolute error is

$$|x(t) - \bar{x}_d(t)| = \left| (x_0 - A) e^{-\omega_r t} \left(e^{-\int_0^t f(s)ds} - e^{-\omega_{t_s} \sum f(t_j)} \right) \right|. \quad (35)$$

Since $f(\cdot) \geq 0$ (non-negative activation function), both exponential terms $e^{-\int_0^t f(s)ds}$ and $e^{-\omega_{t_s} \sum f(t_j)}$ are bounded within $(0, 1]$. The maximum possible difference between any two values in this interval is 1, as demonstrated by the extreme case

$$\sup_{a, b \in (0, 1]} |e^{-a} - e^{-b}| = \lim_{a \rightarrow 0^+} |1 - e^{-b}| \leq 1 \quad (36)$$

thus

$$\left| e^{-\int_0^t f(s)ds} - e^{-\omega_{t_s} \sum f(t_j)} \right| \leq 1 \quad (37)$$

substituting into the error expression yields

$$|x(t) - \bar{x}_d(t)| \leq |x_0 - A| e^{-\omega_r t} \cdot 1 = |x_0 - A| e^{-\omega_r t}. \quad (38)$$

E. Extended Theory

1) *Theorem 2: Nonuniform Sampling Extension:* Given an LTC system governed by the IVP in (4) with single-dimensional exogenous input $\mathbf{I}(t)$, consider nonuniform sampling times $0 = t_0 < t_1 < \dots < t_n = t$. Define variable intervals $\Delta t_i = t_i - t_{i-1}$. The closed-form approximate solution is

$$\begin{aligned} \tilde{x}_d(t) &= (x_0 - A) \\ &\odot \exp \left(-\omega_r t - \sum_{i=1}^n \omega_{\Delta t_i} f(I(t_i), x(t_i), \theta) \right) \\ &\odot f(-I(t), x(t), \theta) + A \end{aligned} \quad (39)$$

where the interval correction factor $\omega_{\Delta t_i}^{(1 \times 1)}$ is

$$\omega_{\Delta t_i} = \exp(\Delta t_i (\ln \Delta t_i - 1)). \quad (40)$$

The approximation error satisfies

$$|x(t) - \tilde{x}_d(t)| \leq |x_0 - A| e^{-\omega_r t}. \quad (41)$$

The detailed derivation process is provided in Appendix A.

2) *Theorem 3: Multivariate Extension:* Given an LTC system with D cells governed by the IVP in (4)

$$\frac{d\mathbf{x}}{dt} = -[\omega_r + f(\mathbf{x}, \mathbf{I}, \theta)] \odot \mathbf{x}(t) + \mathbf{A} \odot f(\mathbf{x}, \mathbf{I}, \theta) \quad (42)$$

where $\mathbf{x}(t) \in \mathbb{R}^D$, $\mathbf{I}(t) \in \mathbb{R}^m$. Under nonuniform sampling $0 = t_0 < t_1 < \dots < t_n = t$, the closed-form approximate solution for the k th cell ($k = 1, \dots, D$) is

$$\begin{aligned} \tilde{x}^{(k)}(t) &= \left(x_0^{(k)} - A^{(k)} \right) \\ &\odot \exp \left(-\omega_r^{(k)} t - \sum_{i=1}^n \omega_{\Delta t_i} f^{(k)}(\mathbf{I}(t_i), \mathbf{x}(t_i), \theta) \right) \\ &f^{(k)}(-\mathbf{I}(t), \mathbf{x}(t), \theta) + A^{(k)} \end{aligned} \quad (43)$$

with $\Delta t_i = t_i - t_{i-1}$, $\omega_{\Delta t_i} = \exp(\Delta t_i (\ln \Delta t_i - 1))$. The approximation error satisfies

$$|x^{(k)}(t) - \tilde{x}^{(k)}(t)| \leq |x_0^{(k)} - A^{(k)}| e^{-\omega_r^{(k)} t}. \quad (44)$$

The detailed derivation process is provided in Appendix B.

3) *Theorem 4: Multilayer Coupled Extension:* Consider L coupled LTC layers indexed by $l = 1, 2, \dots, L$. The l th layer's dynamics are governed by

$$\begin{aligned} \frac{d\mathbf{x}^{(l)}}{dt} &= -[\omega_r^{(l)} + f^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}, \theta^{(l)})] \\ &\odot \mathbf{x}^{(l)}(t) + \mathbf{A}^{(l)} \odot f^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}, \theta^{(l)}) \end{aligned} \quad (45)$$

where $\mathbf{x}^{(0)}(t) = \mathbf{I}(t)$, and $\mathbf{x}^{(l-1)}(t)$ is the state vector of layer $l - 1$. Under nonuniform sampling $0 = t_0 < t_1 < \dots < t_n = t$, the closed-form approximate solution for layer l at time t is

$$\begin{aligned} \tilde{\mathbf{x}}^{(l)}(t) &= \left(\mathbf{x}_0^{(l)} - \mathbf{A}^{(l)} \right) \\ &\odot \exp \left(-\omega_r^{(l)} t - \sum_{i=1}^n \omega_{\Delta t_i} f^{(l)}(\tilde{\mathbf{x}}^{(l)}(t_i), \tilde{\mathbf{x}}^{(l-1)}(t_i), \theta^{(l)}) \right) \\ &\odot f^{(l)}(-\tilde{\mathbf{x}}^{(l-1)}(t), \tilde{\mathbf{x}}^{(l)}(t), \theta^{(l)}) + \mathbf{A}^{(l)} \end{aligned} \quad (46)$$

with $\omega_{\Delta t_i} = \exp(\Delta t_i (\ln \Delta t_i - 1))$, $\Delta t_i = t_i - t_{i-1}$, and $\tilde{\mathbf{x}}^{(l-1)}$ being the approximate solution of layer $l - 1$ from Theorem 4

(for $l \geq 2$) or Theorem 3 (for $l = 1$). The approximation error satisfies

$$\begin{aligned} & \|\mathbf{x}^{(l)}(t) - \tilde{\mathbf{x}}^{(l)}(t)\|_{\infty} \\ & \leq \mathcal{C}^{(l)} e^{-\lambda^{(l)} t} + \gamma^{(l)} \max_{s \leq t} \|\mathbf{x}^{(l-1)}(s) - \tilde{\mathbf{x}}^{(l-1)}(s)\|_{\infty} \end{aligned} \quad (47)$$

where $\mathcal{C}^{(l)} = \max_k |x_0^{(l,k)} - A^{(l,k)}|$, $\lambda^{(l)} = \min_k \omega_{\tau}^{(l,k)}$, and $\gamma^{(l)}$ is the Lipschitz constant of $f^{(l)}$ with respect to $\mathbf{x}^{(l-1)}$. The detailed derivation process is provided in Appendix C.

IV. MODEL BUILDING

Although we have derived different conclusions applicable to nonuniform sampling, multivariate, and multilayer architectures, in the neural network construction below, only the state equations of univariate, single-layer architectures are required. Since Δt_i refers to the sampling interval at each time step, before a batch of data is input, $\omega_{\Delta t_i}$ has already been generated and can be used as a normal variable for multidimensional parallel input. By utilizing matrix parallel computing to calculate the correction values for each sampling interval, this will be equivalent to the conclusion form derived from uniform sampling. Therefore, in the subsequent derivations, the closed-form solution structure of (16) is used for neural network derivations.

In addition, since the inputs $x(t)$ and $I(t)$ received by each neural network instance are concatenated, the formula is converted as follows:

$$\text{concat}(I(t), x(t)) \rightarrow I(t). \quad (48)$$

A. DFA-CfN for Tasks Which Optimized for General Sequential

We can construct (16) into a neural network model suitable for large-scale training: DFA-CfN, the overall framework is shown in Fig. 2. Similar to the pioneering work of CfC, this study represents the solution using the following formula:

$$x(t) = B \odot e^{-\omega_{\tau} t - \omega_{t_s} \sum_{j=1}^n f(I(t_j))} \odot f(-I(t), \theta) + A \quad (49)$$

where the constant coefficients are represented as B and are constructed using the network $b(\cdot)$. Constant matrices A are represented using neural networks $h(\cdot)$. For the time decay term, although $\omega_{t_s} \sum_{j=1}^n f(I(t_j))$ is bounded and does not cause the exponential term to decay rapidly, the time constant term $\omega_{\tau} t_i$ in the exponent can lead to rapid decay as time increases, causing the state solution to quickly converge to a constant A . Therefore, consistent with the foundational CfC work, this study replaces the exponential decay term with a smoother transition process using the inverse sigmoid nonlinear activation function $\sigma(\cdot)$ and introduces a gating mechanism. This performs $b(\cdot) \odot \sigma(\cdot)$ and $h(\cdot) \odot (1 - \sigma(\cdot))$ on $b(\cdot)$ and $h(\cdot)$, respectively. Since the sigmoid function places ω_{t_s} in the denominator, it transforms into

$$\omega_{t_s} = e^{t_s(1 - \ln t_s)}. \quad (50)$$

In addition, for the terms $-\omega_{\tau} t_i - \omega_{t_s} \sum_{j=1}^n f(I(t_j))$ and $f(-I(t_i), \theta)$, the two terms are combined and replaced with a dual-head network $a(\cdot) + c(\cdot)$, representing the decay term

and the nonlinear function term in the original solution. The gating factor T_{interp} is obtained by multiplying $a(\cdot)$ with the interval correction factor ω_{t_s}

$$T_{interp} = a(\cdot) \odot \omega_{t_s} + c(\cdot). \quad (51)$$

Since $a(\cdot)$, $b(\cdot)$, $c(\cdot)$, and $h(\cdot)$ all receive the aggregated mixed state from the sequence's hidden state and exogenous inputs, a shared prenetwork $I(\cdot)$ is set up to realize the input aggregation. Thus, the network forms a structure with $I(\cdot)$ as the main trunk and $a(\cdot)$, $b(\cdot)$, $c(\cdot)$ and $h(\cdot)$ as branches

$$x(t) = \sigma(T_{interp}) \odot b(\cdot) + (1 - \sigma(T_{interp})) \odot h(\cdot). \quad (52)$$

Specifically, to retain the memory effect of $\sum_{j=1}^n f(I(t_j))$, this study designs memory variables M and a memory list L_M . At each time step, the computed T_{interp} is accumulated and summed to M , and the T_{interp} in (52) is replaced with the M . At the initial stage of the recursive connection, the value of M is initialized to 0. Since it is an additive operation, it does not affect the gradient calculation during backpropagation. Instead, it provides an additional method to directly calculate the gradient of the output with respect to the state at each specific time step, which helps stabilize the gradient. At this point, $x(t)$ can be expressed as

$$x(t) = \sigma(M) \odot b(\cdot) + (1 - \sigma(M)) \odot h(\cdot). \quad (53)$$

Since T_{interp} is merged from the decay term and $f(-I(t_i), \theta)$, it contains some feature information of the neural nonlinear aggregation. Establish L_M to store T_{interp} at each step, and after the recursive connection ends, perform temporal concatenation on the elements in the list to obtain the memory sequence $F^{(B \times T \times H)}$ of each temporal decay state. This is connected with the sequence state variable $R^{(B \times T \times H)} = \{x_1, x_2, \dots, x_T\}$ obtained from the recursive connection to get the final output sequence $R_O^{(B \times T \times H)}$.

B. Prereconstruct DFA-CfN for Tasks Which Specialized for Decomposable Sequential Inputs

For (16), extract and expand $\omega_{t_s} \sum_{j=1}^n f(I(t_j))$

$$\omega_{t_s} \sum_{j=1}^n f(t_j) = \omega_{t_s} f(I(t_1)) + \omega_{t_s} f(I(t_2)) + \dots + \omega_{t_s} f(I(t_n)) \quad (54)$$

reconstruct the weights at each integration node and denote them as ω_i . At this point, the above formula becomes a linear neural network with output $y^{(1 \times 1)}$

$$y^{(1 \times 1)} = \omega_1 f(I(t_1)) + \omega_2 f(I(t_2)) + \dots + \omega_n f(I(t_n)) \quad (55)$$

which aggregates the function values at various integration nodes. By expanding the output dimension of the linear network to n , we have

$$y^{(n \times 1)} = \mathbf{W} \mathbf{f} \quad (56)$$

where $\mathbf{f} = [f(I(t_1)), \dots, f(I(t_n))]^T \in \mathbb{R}^{n \times 1}$ and $\mathbf{W} \in \mathbb{R}^{n \times n}$ is a weight matrix. $y^{(n \times 1)}$ can be seen as the reconstructed input sequence obtained by fusing information from each node. Each new function value at a node is constructed from the function values at any original points, thus reducing information loss for

sequences with fixed sampling resolution. However, the model parameters will grow as $O(n^2)$. Considering the parameter explosion problem when the input sequence is too long, the model adopts a sequence blocking operation. First, the input sequence $I^{(T,N)}$ is divided into blocks with temporal dependencies, where N is the number of features, independent of the time length

$$I^{(T,N)} \rightarrow I^{(N_m, T_m, N)} \quad (57)$$

where $T = N_m * T_m$. When the module performs the reconstruction operation, it uses two linear layers, calculated with N_m and T_m as input and output dimensions, respectively, get the output $I_1^{(N_m \times T_m \times N)}$ and $I_2^{(T_m \times N_m \times N)}$

$$I_1 = \mathbf{W}_{N_m} \mathbf{f}_{N_m} \quad (58)$$

$$I_2 = \mathbf{W}_{T_m} \mathbf{f}_{T_m} \quad (59)$$

where $\mathbf{f}_{N_m} \in \mathbb{R}^{N_m \times T_m \times N}$, $\mathbf{W}_{N_m} \in \mathbb{R}^{N_m \times N_m}$, $\mathbf{f}_{T_m} \in \mathbb{R}^{T_m \times N_m \times N}$, and $\mathbf{W}_{T_m} \in \mathbb{R}^{T_m \times T_m}$.

To mitigate the impact of the blocking operation on the overall sequence features, a 1-D convolution is added before blocking to homogenize the temporal features of the sequence. In addition, multiple decoupled sequences based on prior knowledge are weighted and summed to obtain the final reconstructed input $\hat{I}^{(T,N)}$. During this process, the sequence blocking needs to follow:

$$(N_m^2 - 1)(T_m^2 - 1) > 1 \quad (60)$$

to achieve the effect of reducing the number of parameters.

By placing this blocking reconstruction module such as Fig. 3 in front of DFA-CfN, we obtain the PRDFA-CfN, suitable for decomposable sequences.

V. EXPERIMENT

A. Experimental Settings

To validate the improvements of DFA-CfN and PRDFA-CfN on approximation errors, this study conducted baseline tests on four classical modeling tasks (human pose recognition, sentiment analysis of textual data, medical time series prediction, and robotic kinematics modeling) and performed comparative experiments with general time series models on two engineering tasks (multienergy loads forecasting for IESs and carbon content remote sensing sequence prediction). In addition, to verify the effectiveness of key components in both network architectures, three variant networks were proposed for DFA-CfN and PRDFA-CfN, respectively: 1) original closed-form solution network derived from equations (-S); 2) network without gating mechanisms (-noGate); and 3) network embedding RNN states in temporal steps (-mmRNN). These variants remain grounded in the theoretical framework of this research, their experimental results are presented alongside comparative benchmarks. In this study, experiments that measured result stability used multiple naturally generated random numbers as random seeds, while experiments that did not measure stability used 2020 as the random seed. Experiments were conducted using PyTorch on an Intel i7-12700 CPU and NVIDIA RTX 3090 GPU. Data and code are available at <https://github.com/nuist-xf/DFA-CfNs>

TABLE I

PERFORMANCE COMPARISON OF PHYSICAL DYNAMICS MODELING

Model	MSE $\pm$ STD	Time per epoch (min)
ODE-RNN	1.904 $\pm$ 0.061	0.79
CT-RNN	1.198 $\pm$ 0.004	0.91
AugmentedLSTM	1.065 $\pm$ 0.006	0.10
CT-GRU	1.172 $\pm$ 0.011	0.18
RNN-Decay	1.406 $\pm$ 0.005	0.16
Bi-directional RNN	1.071 $\pm$ 0.009	0.39
GRU-D	1.090 $\pm$ 0.034	0.11
PhasedLSTM	1.063 $\pm$ 0.010	0.25
GRU-ODE	1.051 $\pm$ 0.018	0.56
CT-LSTM	1.014 $\pm$ 0.014	0.31
ODE-LSTM	0.883 $\pm$ 0.014	0.29
coRNN	3.241 $\pm$ 0.215	0.18
Lipschitz RNN	1.781 $\pm$ 0.013	0.17
LTC	0.662 $\pm$ 0.013	0.78
Transformer	0.761 $\pm$ 0.032	0.80
CfC	0.643 $\pm$ 0.006	0.08
CfC-mmRNN	0.617 $\pm$ 0.006	0.34
CfC-noGate	0.650 $\pm$ 0.008	0.21
Cf-S	0.948 $\pm$ 0.009	0.12
DFA-CfN	0.500 $\pm$ 0.003	0.10
DFA-CfN-mmRNN	0.503 $\pm$ 0.003	0.31
DFA-CfN-noGate	0.572 $\pm$ 0.004	0.22
DFA-CfN-S	0.737 $\pm$ 0.010	0.14
PRDFA-CfN	0.190 $\pm$ 0.027	0.11
PRDFA-CfN-mmRNN	0.233 $\pm$ 0.043	0.32
PRDFA-CfN-noGate	0.241 $\pm$ 0.050	0.22
PRDFA-CfN-S	–	–

Note: Bold models indicate proposed architectures. Blue text indicates second-best performance, red text shows optimal results. MSE is Mean Squared Error. All values are reported as mean $\pm$ standard ($n = 5$) deviation.

B. Four Types of Basic Continuous Time Series Modeling Tasks

1) *Physical Dynamics Modeling*: The Walker2D dataset [25], generated using the MuJoCo physics engine, provides high-fidelity kinematic simulations of bipedal locomotion. This benchmark dataset contains multijoint dynamic measurements (including joint angles, angular velocities, forces, and torques) sampled at irregular intervals, specifically designed to evaluate physical dynamic system modeling under nonuniform observational conditions. It serves as a critical testbed for robotic motion prediction tasks, particularly for validating the robustness and generalization capabilities of dynamic modeling approaches in reconstructing complex musculoskeletal interactions during ambulatory movements.

The test results presented in Table I demonstrate significant performance improvements of our proposed models. The DFA-CfN architecture exhibits superior accuracy compared to the

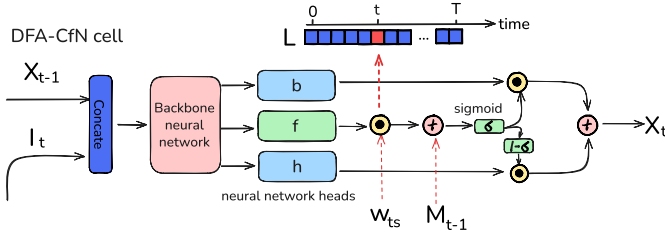


Fig. 2. Architecture of DFA-CfN cell.

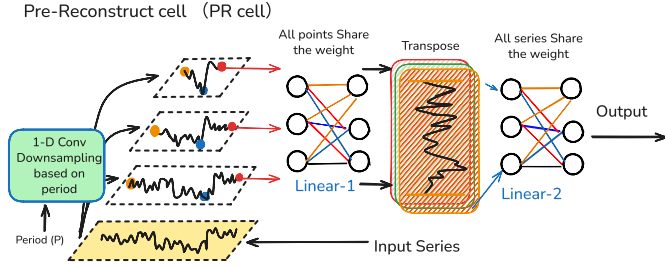


Fig. 3. Architecture of prereconstruct module cell.

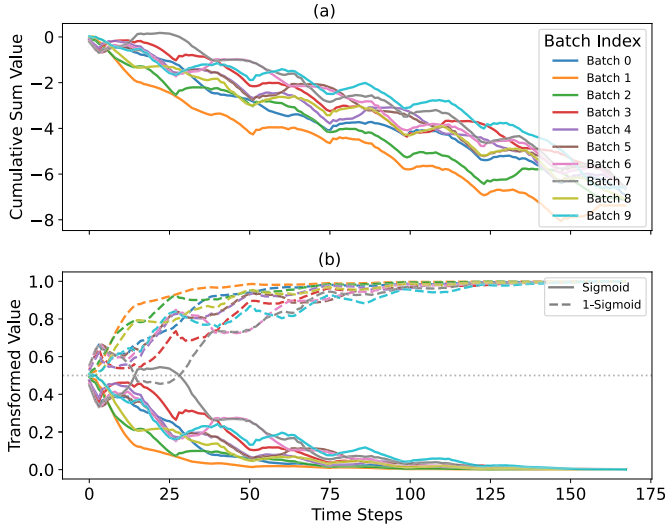


Fig. 4. DFA-CfN dynamic feature accumulation and gating curve diagram on the IEs test set. $\sigma(\cdot)$ mainly plays a role at earlier times (when t is small), controlling the contribution of function $g(\cdot)$; $1 - \sigma(\cdot)$ dominates at later times (when t is large), controlling the contribution of function $h(\cdot)$. (a) Original cumulative sums. (b) Sigmoid transformations.

previous state-of-the-art CfC model across all experimental configurations, with consistent error reduction patterns. More notably, the PRDFA-CfN achieves a mean squared error (mse) of 0.190, outperforming all baseline models by a significant margin (69.2% error reduction compared to the second-best CfC-mmRNN: 0.617). This substantial performance gap quantitatively verifies our models' enhanced proficiency in modeling irregularly sampled physical dynamics, particularly in capturing nonlinear interactions within complex multibody systems under nonuniform temporal observations.

The reason for the lack of results in PRDFA-CfN-S is the disappearance of the gradient. Although it achieves noise signal filtering and enhances the amplitude of mid-to-low

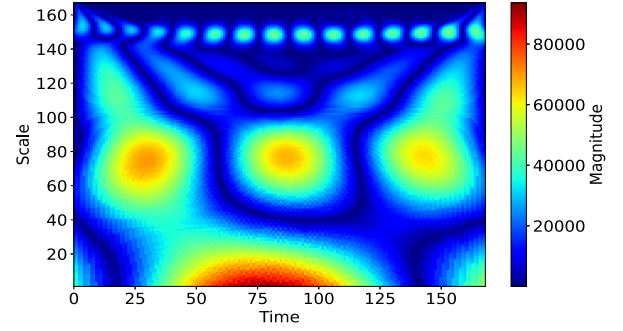


Fig. 5. Multiscale wavelet transform results for the original sequence. Before reconstruction, there are local noises with 90 000 energy in the ultrahigh frequency part and weak information in the intermediate frequency part, which is between 60 000 and 70 000 energy values.

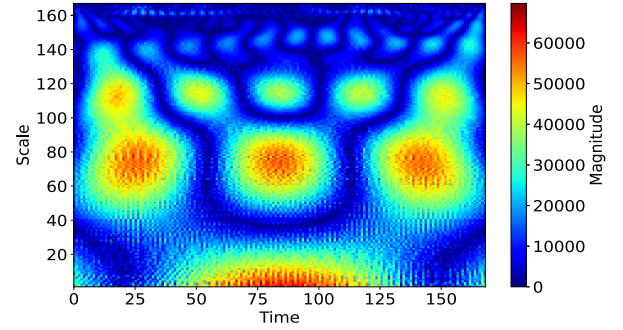


Fig. 6. Multiscale wavelet transform results for raw resolution reconstructed sequences. The energy value of the ultrahigh frequency noise is weakened, while the energy of the medium and low frequency information is enhanced.

frequency signals, this enhancement mechanism, when applied to the DFA-CfN-S network, which is sensitive to signal amplitude, accelerates the accumulation of the functional features of the E-index term. The value obtained from the E-exponential term rapidly decays to 0 after a short time step, causing the gradient to vanish. However, it is worth noting that the PRDFA-CfN-S/DFA-CfN-S models are only used as ablation models to demonstrate the performance of the network directly derived from the original theoretical expression, while also serving as a comparison for the DFA-CfN model and the PRDFA-CfN model.

2) *Human Activity Recognition*: The human activity dataset [26] comprises multimodal time-series data collected from five subjects performing seven ambulatory states (walking, sitting, lying, etc.), containing 6554 multivariate time series with nonuniform temporal sampling. Each observation sequence spans 211 temporal samples captured through triaxial accelerometers positioned at strategic anatomical locations (waist, chest, and ankles), generating 12 kinematic features (3-D positional coordinates per sensor). This benchmark dataset facilitates the development and evaluation of temporal classification models for seven distinct human activity categories, particularly focusing on irregularly sampled motion pattern recognition in wearable computing applications. The proposed model outputs a probability distribution on this dataset, generates categories using the argmax function, and then uses the

TABLE II

PERFORMANCE COMPARISON OF HUMAN ACTIVITY RECOGNITION (%)

Model	Accuracy $\pm$ STD	Time per epoch (min)
RNN-Impute	79.50 $\pm$ 0.80	0.38
RNN- Δt	79.50 $\pm$ 0.80	0.45
RNN-Decay	80.00 $\pm$ 1.00	0.39
GRU-D	80.60 $\pm$ 0.70	0.15
RNN-VAE	34.30 $\pm$ 4.00	2.63
Latent-ODE-RNN	83.50 $\pm$ 1.00	7.71
ODE-RNN	82.90 $\pm$ 1.60	3.15
Latent-ODE-ODE	84.60 $\pm$ 1.30	8.49
CfC	84.87 $\pm$ 0.42	0.084
CfC-mmRNN	85.97 $\pm$ 0.25	0.128
CfC-noGate	85.57 $\pm$ 0.34	0.093
Cf-S	87.04 $\pm$ 0.47	0.097
DFA-CfN	84.94 $\pm$ 0.31	0.087
DFA-CfN-mmRNN	85.98 $\pm$ 0.49	0.134
DFA-CfN-noGate	86.07 $\pm$ 0.21	0.099
DFA-CfN-S	87.86 $\pm$ 0.56	0.101
PRDFA-CfN	86.18 $\pm$ 0.37	0.149
PRDFA-CfN-mmRNN	86.73 $\pm$ 0.22	0.198
PRDFA-CfN-noGate	87.60 $\pm$ 0.35	0.161
PRDFA-CfN-S	88.75 $\pm$ 0.12	0.168

Note: **Bold** models indicate proposed architectures. Blue text indicates second-best performance, red text shows optimal results. All values are reported as mean $\pm$ standard ($n = 5$) deviation.

cross-entropy loss function for backpropagation to train the classifier.

As shown in Table II, the proposed PRDFA-CfN-S model achieves a classification accuracy of 88.75%, significantly outperforming all baseline models (the second-best model DFA-CfN-S: 87.86% $\pm$ 0.56%) with the smallest standard deviation of 0.12%. Specifically, the model addresses the multiscale spatial-temporal coupling of motion patterns in wearable sensor data through prereconstruct method, validated the validity of the model.

3) *PhysioNet Challenge*: The PhysioNet Challenge 2012 dataset is widely utilized in medical time series research. Derived from records of 12000 intensive care unit (ICU) patients, this dataset comprises adult patients admitted to cardiac, medical, surgical, and trauma ICUs due to diverse etiologies, while excluding cases with ICU stays shorter than 48 hours. The classification objective of the PhysioNet dataset is to determine whether patients died during their hospital stay, which is a binary classification task (survival/death). Following the same experimental setup as prior work [12], this study obtained the final test results in Table III.

The DFA series models demonstrate superior performance on the PhysioNet Challenge 2012 dataset. Among them, DFA-CfN-mmRNN achieves the highest AUC of 0.8503 $\pm$ 0.001, followed by DFA-CfN (0.8493 $\pm$ 0.001) and DFA-CfN-noGate (0.8447 $\pm$ 0.002). Notably, DFA-CfN-S improves

TABLE III

PERFORMANCE COMPARISON OF PHYSIONET CHALLENGE

Model	AUC $\pm$ STD	Time per epoch (min)
RNN-Impute	0.764 $\pm$ 0.016	0.5
RNN- Δt	0.787 $\pm$ 0.014	0.5
RNN-Decay	0.807 $\pm$ 0.003	0.7
GRU-D	0.818 $\pm$ 0.008	0.7
Phased-LSTM	0.836 $\pm$ 0.003	0.3
IP-Nets	0.819 $\pm$ 0.006	1.3
SeFT	0.795 $\pm$ 0.015	0.5
RNN-VAE	0.515 $\pm$ 0.040	2.0
Latent-ODE-RNN	0.781 $\pm$ 0.018	6.7
ODE-RNN	0.833 $\pm$ 0.009	16.5
Latent-ODE-ODE	0.829 $\pm$ 0.004	22.0
LTC	0.647 $\pm$ 0.010	0.5
CfC	0.839 $\pm$ 0.002	0.1
CfC-mmRNN	0.834 $\pm$ 0.006	0.2
CfC-noGate	0.840 $\pm$ 0.003	0.1
Cf-S	0.643 $\pm$ 0.018	0.1
DFA-CfN	0.8493 $\pm$ 0.001	0.1
DFA-CfN-mmRNN	0.8503 $\pm$ 0.001	0.2
DFA-CfN-noGate	0.8447 $\pm$ 0.002	0.1
DFA-CfN-S	0.8301 $\pm$ 0.007	0.1
PRDFA-CfN	—	—
PRDFA-CfN-mmRNN	—	—
PRDFA-CfN-noGate	—	—
PRDFA-CfN-S	—	—

Note: **Bold** models indicate proposed architectures. Blue text indicates second-best performance, red text shows optimal result. "—" denotes unavailable data. AUC is the area under the curve. All values are reported as mean $\pm$ standard ($n = 5$) deviation.

the AUC by approximately 29% compared to Cf-S. Due to the variable sequence lengths of input data in the PhysioNet Challenge 2012 dataset, the PRDFA series models (which rely on fixed decomposition hyperparameters for sequence PR) were unable to produce experimental results on this dataset. This limitation highlights the challenges in applying fixed-parameter decomposition methods to irregularly sampled clinical time series.

4) *Sentiment Analysis Using IMDB*: The IMDB sentiment analysis dataset [27] is widely used in natural language processing (NLP) as a benchmark for sentiment classification tasks. Derived from the IMDB, it comprises 50000 movie reviews labeled as positive or negative for binary sentiment classification. The dataset is evenly split into 25000 training and 25000 testing samples, with balanced distribution of positive and negative labels (50% each). This equilibrium enhances model training and evaluation by minimizing class bias. Reviews exhibit variable-length texts, typically spanning hundreds of words, which challenges models' ability to process long sequences effectively. In order to convert the text, this study used an embedding layer.

TABLE IV

PERFORMANCE COMPARISON OF IMDB SENTIMENT ANALYSIS DATASET

Model	Accuracy $\pm$ STD (%)
HiPPO-LagT	88.0 $\pm$ 0.2
HiPPO-LegS	88.0 $\pm$ 0.2
LMU	87.7 $\pm$ 0.1
LSTM	87.3 $\pm$ 0.4
GRU	86.2 $\pm$ n/a
ReLU GRU	84.8 $\pm$ n/a
Skip LSTM	86.6 $\pm$ n/a
expRNN	84.3 $\pm$ 0.3
Vanilla RNN	67.4 $\pm$ 7.7
coRNN	86.7 $\pm$ 0.3
LTC	61.8 $\pm$ 6.1
CfC	85.9 $\pm$ 0.9
CfC-mmRNN	88.3 $\pm$ 0.1
CfC-noGate	87.5 $\pm$ 0.1
Cf-S	81.7 $\pm$ 0.5
DFA-CfN	87.6 $\pm$ 0.1
DFA-CfN-mmRNN	88.1 $\pm$ 0.2
DFA-CfN-noGate	88.0 $\pm$ 0.1
DFA-CfN-S	—
PRDFA-CfN	88.2 $\pm$ 0.1
PRDFA-CfN-mmRNN	88.5 $\pm$ 0.1
PRDFA-CfN-noGate	88.5 $\pm$ 0.07
PRDFA-CfN-S	—

Note: **Bold** models indicate proposed architectures. Blue text indicates second-best performance, red text shows optimal result. "—" denotes unavailable data, "n/a" indicates non-applicable standard deviation. All values are reported as mean $\pm$ standard ($n = 5$) deviation.

The experimental results in Table IV show that among the proposed new architectures, PRDFA-CfN-mmRNN and PRDFA-CfN-noGate achieve the highest accuracy of 88.5% (with standard deviations of 0.1% and 0.07%, respectively), performing equally optimally, demonstrating the advantage of the PR mechanism in handling long-text sequence tasks. The second-best model is CfC-mmRNN (88.3% $\pm$ 0.1%). Meanwhile, due to the variable-length text characteristics of the IMDB dataset (where reviews typically span hundreds of words), gradient vanishing occurred during inference for DFA-CfN-S and PRDFA-CfN-S, which aligns with previous theoretical deductions. This phenomenon arises because when the accumulative sequence values become excessively large, the exponential terms rapidly decay to zero, leading to gradient disappearance. These findings quantitatively validate the architectural limitations in modeling long-range dependencies within variable-length textual sequences.

C. Two Types of Engineering Time Series Tasks

1) *XCO₂ Remote Sensing Sequence Prediction:* The multi-source XCO₂ concentration prediction dataset holds significant

TABLE V

PERFORMANCE COMPARISON OF XCO₂ REMOTE SENSING SEQUENCE PREDICTION

Model	R ²	MAE	RMSE	MAPE (%)
CNN	0.8797	0.6324	0.9043	0.15
GRU	0.8929	0.5900	0.8531	0.14
LSTM	0.8606	0.7018	0.9733	0.17
Linear	0.9210	0.4953	0.6836	0.11
Dlinear	0.9204	0.5063	0.7064	0.12
Transformer	0.9201	0.5136	0.7033	0.12
iTransformer	0.9315	0.5162	0.6510	0.12
Informer	0.9193	0.5625	0.7069	0.13
Autoformer	0.9329	0.5082	0.6445	0.12
TiDE	0.9283	0.5086	0.6832	0.12
FiTS	0.9307	0.5131	0.6554	0.12
PatchTST	0.9422	0.4401	0.6134	0.11
CfC	0.9591	0.4422	0.6613	0.11
CfC-mmRNN	0.9530	0.4683	0.7090	0.11
CfC-noGate	0.9465	0.4954	0.7570	0.12
Cf-S	0.9437	0.5140	0.7763	0.12
DFA-CfN	0.9654	0.4185	0.6084	0.09
DFA-CfN-mmRNN	0.9611	0.4286	0.6459	0.10
DFA-CfN-noGate	0.9678	0.4097	0.5878	0.09
DFA-CfN-S	0.9337	0.5489	0.8423	0.13
PRDFA-CfN	0.9580	0.4467	0.6711	0.10
PRDFA-CfN-mmRNN	0.9107	0.6463	0.9775	0.15
PRDFA-CfN-noGate	0.9679	0.4112	0.5855	0.09
PRDFA-CfN-S	0.9508	0.4839	0.8259	0.11

Note: **Bold** models indicate proposed architectures. Blue text indicates second-best performance, red text shows optimal result. R-squared (R²), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE).

value in environmental remote sensing and carbon flux studies, serving as a critical benchmark for regional carbon sink/source inversion and carbon neutrality assessment. This dataset integrates multidimensional observations, including ERA5 reanalysis meteorological data from the European Center for Medium-Range Weather Forecasts (ECMWF), MODIS-derived normalized difference vegetation index (NDVI), SRTM digital elevation model (DEM), CLCD30-m land-use classification, and socioeconomic factors, to construct spatial-temporally continuous feature inputs for XCO₂ concentration prediction. Spanning the Yangtze River Delta region of China from 2016 to 2021, the dataset contains 2 850 390 valid spatial-temporal data points at 16-day temporal resolution. Its comprehensive coverage of seasonal cycles and geographical variations rigorously challenges models' ability to capture nonlinear relationships in complex Earth systems.

The performance comparison of various models in XCO₂ remote sensing sequence prediction is summarized in Table V. The proposed PRDFA-CfN-noGate architecture achieves optimal performance across all four metrics, with the highest

TABLE VI
PERFORMANCE (MAPE) COMPARISON OF MULTIENERGY LOADS FORECASTING FOR IES

Model	Electrical Load				Cooling Load				Heating Load			
Steps	24	48	72	96	24	48	72	96	24	48	72	96
Neural CDE [28]	6.691	8.026	8.514	9.247	9.02	14.248	13.364	15.272	5.918	7.200	7.749	7.981
ANCDE [29]	6.016	7.213	8.046	8.573	8.243	10.49	12.78	14.272	4.72	5.704	6.58	7.15
Crossformer	5.384	6.944	8.042	8.105	7.743	11.105	13.025	13.449	4.757	6.097	6.654	7.684
Dlinear	5.698	6.967	7.802	8.420	7.521	10.274	12.520	14.241	4.586	5.570	6.388	6.998
FiLM	6.287	6.941	7.750	8.351	7.633	10.310	12.488	14.150	4.664	5.617	6.461	7.089
FITS	5.568	6.832	7.799	8.528	7.466	10.143	12.318	14.100	4.614	5.554	6.296	6.836
iTransformer	5.464	6.803	7.760	8.327	8.413	11.411	13.180	14.508	4.866	5.847	6.593	7.369
NLinear	5.716	7.013	7.879	8.519	7.524	10.310	12.536	14.260	4.599	5.576	6.368	6.957
PatchTST	5.293	6.727	7.717	8.555	7.976	10.621	13.942	14.674	4.934	5.999	7.042	7.397
SparseTSF	6.912	8.194	8.976	9.588	9.528	12.295	14.472	16.131	5.056	5.984	6.659	7.151
TimeMixer	5.781	6.988	7.778	8.040	7.673	10.194	11.964	13.322	4.623	5.574	6.226	6.817
TimesNet	7.040	8.280	8.721	10.978	7.902	10.869	11.786	13.888	5.404	6.321	6.696	7.073
CfC	5.670	6.940	7.620	8.200	7.380	9.940	11.380	13.040	4.510	5.420	6.020	6.500
CfC-mmRNN	5.763	6.980	7.840	8.501	8.147	10.510	11.440	13.067	4.688	5.520	6.160	6.634
CfC-noGate	5.679	6.930	7.760	8.292	7.394	9.980	11.910	13.373	4.493	5.420	6.100	6.609
Cf-S	5.691	6.940	7.800	8.414	7.450	10.100	11.940	13.475	4.526	5.470	6.120	6.673
DFA-CfN	5.380	6.870	7.190	8.142	7.410	10.120	11.990	13.500	4.520	5.600	6.140	6.700
DFA-CfN-mmRNN	5.309	6.600	7.220	7.680	7.678	10.410	12.260	13.650	4.611	5.510	6.250	6.720
DFA-CfN-noGate	5.360	6.700	7.440	8.110	7.330	10.520	11.750	13.450	4.480	5.460	6.090	6.620
DFA-CfN-S	5.760	6.990	7.840	8.470	7.440	10.060	12.010	13.540	4.470	5.410	6.170	6.750
PRDFA-CfN	5.250	6.490	7.310	7.730	7.113	9.520	11.140	12.930	4.472	5.380	5.950	6.460
PRDFA-CfN-mmRNN	5.150	6.660	7.330	7.810	7.190	10.180	12.050	12.340	4.520	5.380	5.870	6.550
PRDFA-CfN-noGate	5.300	6.690	7.340	7.900	7.160	9.460	11.030	11.520	4.510	5.440	5.860	6.370
PRDFA-CfN-S	5.670	6.960	7.760	7.860	7.330	9.940	12.010	13.020	4.520	5.440	6.030	6.590

Note: Bold models indicate neural differential equation variants. Blue = Second-best, red = Optimal. Mean Absolute Percentage Error (MAPE).

R^2 (0.9679), RMSE (0.5855), and MAPE (0.09%), while its MAE (0.4112) ranks second only to DFA-CfN-noGate (0.4097). This validates the effectiveness of the PR temporal decomposition mechanism in extracting features from multisource remote sensing data. Compared with baseline models, traditional temporal architectures (e.g., LSTM: $R^2 = 0.8606$, RMSE = 0.9733) and linear models (Linear: $R^2 = 0.9210$) exhibit significant limitations in modeling nonlinear relationships. The novel DFA/PRDFA series architectures demonstrate 6.8%–7.3% improvement in MAE over the best conventional model (PatchTST: 0.4401) through dynamic feature accumulation mechanisms. All top-performing models (PRDFA-CfN-noGate, DFA-CfN-noGate) maintain MAPE below 0.1%, highlighting their superior percentage error control capability—a critical requirement for precise carbon flux inversion. The experimental results confirm that models integrating temporal decomposition with dynamic feature accumulation mechanisms can effectively capture the

complex coupling relationships between XCO₂ concentrations and meteorological/geographical factors.

2) *Multienergy Loads Forecasting for Ies*: The IES coordinates energy supply, conversion, and storage through coupled electricity-gas-thermal flows, employing energy cascade utilization to optimize operational efficiency. Load demands are categorized by energy type: electrical loads (power consumption of terminal equipment), thermal loads (heating requirements for building climate control and industrial processes), and cooling loads (refrigeration energy in temperature-sensitive applications). This functional segregation addresses heterogeneous load characteristics through multienergy complementarity [30]. This study utilized publicly available operational data from Arizona State University's Tempe Campus (2021–2023) and meteorological data from the National Centers for Environmental Information (NCEI) to construct a preprocessed dataset for electricity, cooling, and heating load forecasting in IES. The dataset was partitioned

TABLE VII
PERFORMANCE (MAPE) OF DIFFERENT BLOCK CONFIGURATIONS FOR MULTIENERGY LOADS FORECASTING IN IES

Number of blocks	Electrical Load				Cooling Load				Heating Load			
Steps	24	48	72	96	24	48	72	96	24	48	72	96
(6,12,24)	5.250	6.490	7.310	7.730	7.113	9.520	11.140	12.930	4.472	5.380	5.950	6.460
(6)	5.343	6.406	7.480	8.068	7.255	9.437	10.872	12.222	4.529	5.332	5.871	6.349
(12)	5.204	6.708	7.767	7.980	7.241	10.155	12.282	12.045	4.507	5.455	6.429	6.493
(24)	5.244	6.522	7.264	8.289	7.371	9.629	10.863	14.697	4.538	5.425	5.955	6.820

Note: Mean Absolute Percentage Error (MAPE) for different block configurations. Optimal values shown in red, second-best in blue.

into training, validation, and test sets with an 8:1:1 ratio, with the final test results summarized in Table VI.

Experimental results demonstrate the significant advantages of the two novel model series in multienergy load forecasting tasks. The PRDFA-CfN series achieves the lowest errors in 72-/96-h predictions for electrical, cooling, and heating loads. Specifically, PRDFA-CfN-noGate reduces the MAPE to 11.52% for 96-h cooling load forecasting, representing reductions of 14.3% and 19.1% compared to Crossformer (13.45%) and DLinear (14.24%), respectively. Through its PR mechanism, this series effectively captures long-term dependencies of energy loads: PRDFA-CfN-mmRNN sets a new accuracy record with 5.15% MAPE for 24-h electrical load prediction, while PRDFA-CfN achieves 6.46% error in 96-h heating load forecasting, outperforming baseline models by over 7%. The DFA-CfN series demonstrates superior performance in 72-h electrical load prediction with 7.19% MAPE, achieving a 6.8% reduction compared to the conventional temporal model PatchTST (7.717%). Its dynamic feature accumulation mechanism exhibits unique strengths in mid-short-term forecasting, approaching PRDFA-CfN's performance in 48-h cooling load (10.12%) and 24-h heating load predictions (4.52%). These results reveal complementary advantages between the two series across long- and short-term forecasting scenarios.

D. Analytical Experiments on the IEs Test Set

1) *Analysis of Different Numbers of Blocks:* A comparative analysis of MAPE performance across different cycle lengths [multicycle combinations (6, 12, 24) versus single cycles (6), (12), (24)] in Table VII reveals that in electricity load forecasting, multicycle combinations achieve optimal performance at a 96-step forecast horizon (MAPE = 7.730), representing a 6.7% improvement over the single cycle (24), thereby validating the effectiveness of multicycle fusion for long-term forecasting; Cold load forecasting exhibits step-length sensitivity in cycle configuration, with the single-cycle (6) performing best at a 48-step length (9.437), while (12) performs best at a 96-step length (12.045). However, the multicycle combination maintains a lead at a 24-step length (7.113); In heat load forecasting, the multicycle combination demonstrated a significant advantage at a 24-step length (4.472), while the single-cycle (6) performed stably in medium- to long-term forecasts. Comprehensive analysis indicates that although different segmentation strategies each exhibit leading

performance, the differences are not significant. Therefore, the model exhibits strong stability in segmentation strategies. In real-world sequence modeling, the multicycle combination demonstrates optimal stability across step lengths.

2) *Visual Analysis of Dynamic Feature Accumulation and Gating:* As illustrated in Fig. 4, the two components ($\sigma(\cdot)$ and $1 - \sigma(\cdot)$) jointly form a gating mechanism: $\sigma(\cdot)$ primarily operates at earlier times (when t is small), governing the contribution of the function $g(\cdot)$; while $1 - \sigma(\cdot)$ dominates at later times (when t is large), controlling the contribution of the function $h(\cdot)$. This mechanism enables the entire network to smoothly interpolate between the two temporal limits of the ODE trajectory: as $t \rightarrow -\infty$, $\sigma(\cdot)$ approaches 1 and $1 - \sigma(\cdot)$ approaches 0, so the network behavior is predominantly determined by $g(\cdot)$; conversely, as $t \rightarrow \infty$, $\sigma(\cdot)$ approaches 0 and $1 - \sigma(\cdot)$ approaches 1, where the network behavior is mainly governed by $h(\cdot)$. This approach allows the model to regulate the influence of historical information by modulating the gating intensity during the learning process.

3) *Thermal Visualization Analysis of PR Module Characteristics:* The PR module performs downsampling based on predefined periodic parameters and reconstructs the input sequence through intraperiod weighted aggregation and interperiod weighted aggregation. To analyze its transformation mechanism, we use the continuous wavelet transform (CWT) combined with the Morlet mother wavelet to perform multiscale analysis on the reconstructed sequence. This method focuses on the different frequency components obtained from the scale parameters in the reconstructed sequence features (e.g., shallow-layer local, high-frequency details versus deep-layer global, low-frequency semantic contours) and utilizes the time-frequency aggregation characteristics and complex phase information of the Morlet wavelet to precisely characterize the transient intensity and oscillation patterns of feature responses. This aids in identifying key activation patterns, transient feature correlations, and cross-scale information fusion mechanisms during the layer-wise propagation process, providing a new perspective for understanding internal network representations. Fig. 5 represents the original sequence, and Fig. 6 represents the reconstructed sequence. In Fig. 5, the PR sequence exhibits local noise in the ultrahigh-frequency band, with energy exceeding 90 000, while the mid-frequency band shows weaker information, with energy values ranging from 60 000 to 70 000. In Fig. 6, the energy values in the

ultrahigh-frequency band of the reconstructed sequence are attenuated to approximately 60 000, effectively suppressing ultrahigh-frequency noise. In addition, Fig. 6 fully preserves and enhances the low-frequency spatial information of the original sequence (scale 100–160).

VI. CONCLUSION

This study addresses the limitations of existing CfCs in handling nonuniformly sampled data by proposing a numerical integration approximation theory based on lagrange interpolation method. We rederive the closed-form solution for LTC network state equations and construct two novel architectures: the DFA-CfN and the PRDFA-CfN, along with six variant networks from both series. Theoretical analysis demonstrates the error upper bound of the approximate solution and verifies its consistency with classical CfC models when sampling intervals approach infinitesimal scales. Experiments show that the proposed models outperform existing methods across six temporal tasks, achieving up to 69.2% accuracy improvement and demonstrating exceptional capability in capturing complex patterns. This research establishes a new paradigm for continuous-time modeling and provides innovative solutions for interpretable time series analysis.

VII. CODE AVAILABILITY

The code used to support the findings of this study is available from <https://github.com/nuist-xf/DFA-CfN>

APPENDIX A PROOF OF THEOREM 2

Begin with the exact solution [see (19)]

$$x(t) = (x_0 - A) e^{-\omega_\tau t - \int_0^t f(I(s), x(s), \theta) ds} + A \quad (61)$$

decompose the integral over nonuniform intervals

$$\int_0^t f(s) ds = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} f(I(s), x(s), \theta) ds. \quad (62)$$

On each subinterval $[t_{i-1}, t_i]$, apply Lagrange interpolation at endpoints $\{t_{i-1}, t_i\}$

$$\int_{t_{i-1}}^{t_i} f(s) ds \approx f(t_i) \int_{t_{i-1}}^{t_i} \frac{s - t_{i-1}}{\Delta t_i} ds. \quad (63)$$

Define $g(s) = s - t_{i-1}$. Take the natural logarithm

$$\ln \left(\int_{t_{i-1}}^{t_i} g(s) ds \right) \approx \int_{t_{i-1}}^{t_i} \ln g(s) ds. \quad (64)$$

Solve via integration by parts

$$\int \ln g(s) ds = g(s) \ln g(s) - g(s) \Big|_{t_{i-1}}^{t_i} = \Delta t_i \ln \Delta t_i - \Delta t_i \quad (65)$$

exponentiate to recover the integral

$$\int_{t_{i-1}}^{t_i} g(s) ds = \exp(\Delta t_i \ln \Delta t_i - \Delta t_i) = \omega_{\Delta t_i} \quad (66)$$

substitute into the approximation

$$\int_{t_{i-1}}^{t_i} f(s) ds \approx f(t_i) \omega_{\Delta t_i} \quad (67)$$

sum over all intervals

$$\int_0^t f(s) ds \approx \sum_{i=1}^n f(t_i) \omega_{\Delta t_i} \quad (68)$$

thus $\tilde{x}_d(t)$ follows directly. When $\Delta t_i \equiv t_s$, the correction factor reduces to:

$$\omega_{\Delta t_i} = \exp(t_s (\ln t_s - 1)) = \omega_{t_s}. \quad (69)$$

Error Bound:

Since $0 \leq f \leq 1$:

$$e^{-\int_0^t f ds}, e^{-\sum \omega_{\Delta t_i} f(t_i)} \in [0, 1] \quad (70)$$

their difference is bounded by

$$\left| e^{-\int_0^t f ds} - e^{-\sum \omega_{\Delta t_i} f(t_i)} \right| \leq 1 \quad (71)$$

therefore

$$|x(t) - \tilde{x}_d(t)| \leq |(x_0 - A) e^{-\omega_\tau t} \cdot 1| = |x_0 - A| e^{-\omega_\tau t}. \quad (72)$$

APPENDIX B PROOF OF THEOREM 3

The state equation for the k th cell is derived from (4)

$$\begin{aligned} \frac{dx^{(k)}}{dt} = & -[\omega_\tau^{(k)} + f^{(k)}(\mathbf{x}, \mathbf{I}, \theta)] \\ & \odot x^{(k)}(t) + A^{(k)} f^{(k)}(\mathbf{x}, \mathbf{I}, \theta) \end{aligned} \quad (73)$$

its exact solution is

$$\begin{aligned} x^{(k)}(t) = & \left(x_0^{(k)} - A^{(k)} \right) \\ & \odot e^{-\omega_\tau^{(k)} t - \int_0^t f^{(k)}(\mathbf{I}(s), \mathbf{x}(s), \theta) ds} + A^{(k)}. \end{aligned} \quad (74)$$

Decompose the integral into nonuniform subintervals

$$\int_0^t f^{(k)}(s) ds = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} f^{(k)}(\mathbf{I}(s), \mathbf{x}(s), \theta) ds \quad (75)$$

apply the NIALIM approximation on each $[t_{i-1}, t_i]$

$$\int_{t_{i-1}}^{t_i} f^{(k)}(s) ds \approx f^{(k)}(\mathbf{I}(t_i), \mathbf{x}(t_i), \theta) \omega_{\Delta t_i} \quad (76)$$

summation yields

$$\int_0^t f^{(k)}(s) ds \approx \sum_{i=1}^n \omega_{\Delta t_i} f^{(k)}(\mathbf{I}(t_i), \mathbf{x}(t_i), \theta) \quad (77)$$

substitution gives the closed-form solution.

Error Bound:

Since $0 \leq f^{(k)} \leq 1$

$$\left| e^{-\int_0^t f^{(k)} ds} - e^{-\sum \omega_{\Delta t_i} f^{(k)}(t_i)} \right| \leq 1 \quad (78)$$

thus the error bound holds. When $D = 1$, this reduces to Theorem 2.

APPENDIX C PROOF OF THEOREM 4

The exact solution for layer $l \geq 2$ is

$$\begin{aligned} \mathbf{x}^{(l)}(t) &= \left(\mathbf{x}_0^{(l)} - \mathbf{A}^{(l)} \right) \\ &\odot \exp \left(-\omega_\tau^{(l)} t - \int_0^t f^{(l)}(\mathbf{x}^{(l)}(s), \mathbf{x}^{(l-1)}(s), \theta^{(l)}) ds \right) \\ &\odot f^{(l)}(-\mathbf{x}^{(l-1)}(t), \mathbf{x}^{(l)}(t), \theta^{(l)}) + \mathbf{A}^{(l)} \end{aligned} \quad (79)$$

decompose the integral over nonuniform intervals

$$\int_0^t f^{(l)}(s) ds = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} f^{(l)}(\mathbf{x}^{(l)}(s), \mathbf{x}^{(l-1)}(s), \theta^{(l)}) ds \quad (80)$$

apply the NIALIM approximation on each subinterval $[t_{i-1}, t_i]$

$$\int_{t_{i-1}}^{t_i} f^{(l)}(s) ds \approx f^{(l)}(\mathbf{x}^{(l)}(t_i), \mathbf{x}^{(l-1)}(t_i), \theta^{(l)}) \omega_{\Delta t_i} \quad (81)$$

substitute $\mathbf{x}^{(l)} \approx \tilde{\mathbf{x}}^{(l)}$ and $\mathbf{x}^{(l-1)} \approx \tilde{\mathbf{x}}^{(l-1)}$

$$\int_{t_{i-1}}^{t_i} f^{(l)}(s) ds \approx f^{(l)}(\tilde{\mathbf{x}}^{(l)}(t_i), \tilde{\mathbf{x}}^{(l-1)}(t_i), \theta^{(l)}) \omega_{\Delta t_i} \quad (82)$$

summation yields

$$\int_0^t f^{(l)}(s) ds \approx \sum_{i=1}^n \omega_{\Delta t_i} f^{(l)}(\tilde{\mathbf{x}}^{(l)}(t_i), \tilde{\mathbf{x}}^{(l-1)}(t_i), \theta^{(l)}) \quad (83)$$

substitution into the exact solution gives the closed-form approximation for layer l .

Error Dynamics and Bound:

Define the error $\mathbf{e}^{(l)}(t) = \mathbf{x}^{(l)}(t) - \tilde{\mathbf{x}}^{(l)}(t)$. The exact dynamics are

$$\frac{d\mathbf{x}^{(l)}}{dt} = -\mathbf{\Omega}^{(l)} \odot \mathbf{x}^{(l)}(t) + \mathbf{B}^{(l)}(t) \quad (84)$$

where $\mathbf{\Omega}^{(l)} = \omega_\tau^{(l)} + f^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}, \theta^{(l)})$, $\mathbf{B}^{(l)}(t) = \mathbf{A}^{(l)} \odot f^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}, \theta^{(l)})$. The approximate dynamics are

$$\frac{d\tilde{\mathbf{x}}^{(l)}}{dt} = -\tilde{\mathbf{\Omega}}^{(l)} \odot \tilde{\mathbf{x}}^{(l)}(t) + \tilde{\mathbf{B}}^{(l)}(t) \quad (85)$$

where $\tilde{\mathbf{\Omega}}^{(l)} = \omega_\tau^{(l)} + f^{(l)}(\tilde{\mathbf{x}}^{(l)}, \tilde{\mathbf{x}}^{(l-1)}, \theta^{(l)})$, $\tilde{\mathbf{B}}^{(l)}(t) = \mathbf{A}^{(l)} \odot f^{(l)}(\tilde{\mathbf{x}}^{(l)}, \tilde{\mathbf{x}}^{(l-1)}, \theta^{(l)})$. The error dynamics follow

$$\frac{d\mathbf{e}^{(l)}}{dt} = -\mathbf{\Omega}^{(l)} \odot \mathbf{e}^{(l)}(t) + \mathbf{G}^{(l)}(t) \quad (86)$$

where

$$\begin{aligned} \mathbf{G}^{(l)}(t) &= \left(\mathbf{\Omega}^{(l)} - \tilde{\mathbf{\Omega}}^{(l)} \right) \odot \tilde{\mathbf{x}}^{(l)}(t) \\ &+ \left(\mathbf{B}^{(l)}(t) - \tilde{\mathbf{B}}^{(l)}(t) \right). \end{aligned} \quad (87)$$

By Lipschitz continuity of $f^{(l)}$

$$\left\| \mathbf{\Omega}^{(l)} - \tilde{\mathbf{\Omega}}^{(l)} \right\|_\infty \leq \gamma_1^{(l)} \left\| \mathbf{e}^{(l)} \right\|_\infty + \gamma_2^{(l)} \left\| \mathbf{e}^{(l-1)} \right\|_\infty \quad (88)$$

$$\begin{aligned} \left\| \mathbf{B}^{(l)} - \tilde{\mathbf{B}}^{(l)} \right\|_\infty \\ \leq \left\| \mathbf{A}^{(l)} \right\|_\infty \left(\gamma_1^{(l)} \left\| \mathbf{e}^{(l)} \right\|_\infty + \gamma_2^{(l)} \left\| \mathbf{e}^{(l-1)} \right\|_\infty \right) \end{aligned} \quad (89)$$

where $\gamma_1^{(l)}, \gamma_2^{(l)}$ are Lipschitz constants with respect to $\mathbf{x}^{(l)}$ and $\mathbf{x}^{(l-1)}$. Thus,

$$\left\| \mathbf{G}^{(l)}(t) \right\|_\infty \leq K^{(l)} \left(\left\| \mathbf{e}^{(l)} \right\|_\infty + \gamma^{(l)} \left\| \mathbf{e}^{(l-1)} \right\|_\infty \right) \quad (90)$$

with $K^{(l)} = (\left\| \tilde{\mathbf{x}}^{(l)} \right\|_\infty + \left\| \mathbf{A}^{(l)} \right\|_\infty)$, $\gamma^{(l)} = \gamma_2^{(l)}$. Solving the error ODE

$$\mathbf{e}^{(l)}(t) = e^{-\mathbf{\Lambda}^{(l)} t} \mathbf{e}^{(l)}(0) + \int_0^t e^{-\mathbf{\Lambda}^{(l)}(t-s)} \mathbf{G}^{(l)}(s) ds \quad (91)$$

where $\mathbf{\Lambda}^{(l)} = \text{diag}(\mathbf{\Omega}^{(l)})$. Taking ∞ -norm and bounding

$$\begin{aligned} \left\| \mathbf{e}^{(l)}(t) \right\|_\infty &\leq C^{(l)} e^{-\lambda^{(l)} t} + \int_0^t e^{-\lambda^{(l)}(t-s)} \\ &\left(K^{(l)} \left\| \mathbf{e}^{(l)}(s) \right\|_\infty + K^{(l)} \gamma^{(l)} \left\| \mathbf{e}^{(l-1)}(s) \right\|_\infty \right) ds. \end{aligned} \quad (92)$$

The term $\int_0^t e^{-\lambda^{(l)}(t-s)} \left\| \mathbf{e}^{(l)}(s) \right\|_\infty ds$ is absorbed into the homogeneous solution via Grönwall's inequality. The critical term is

$$\begin{aligned} K^{(l)} \gamma^{(l)} \int_0^t e^{-\lambda^{(l)}(t-s)} \left\| \mathbf{e}^{(l-1)}(s) \right\|_\infty ds \\ \leq \frac{K^{(l)} \gamma^{(l)}}{\lambda^{(l)}} \max_{s \leq t} \left\| \mathbf{e}^{(l-1)}(s) \right\|_\infty \end{aligned} \quad (93)$$

setting $\gamma^{(l)} := K^{(l)} \gamma_2^{(l)} / \lambda^{(l)}$ yields the bound. For bounded error propagation

$$\gamma^{(l)} < 1, \quad \lambda^{(l)} > 0 \quad (94)$$

this ensures exponential decay of interlayer error coupling.

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